Visual Training

Of

Microsoft Visual Studio2015

Web based Applications using ASP.Net with MS Visual C-Sharp

ASP.NET

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Chapter No.4

Giving Beauty to your Web Site using Casecading Style Sheet (CSS)

DAY4

In Today's lecture,

- *What is Cascading Style Sheet (CSS)*
- *Types of CSS*
- *Using the Style Sheet to give consistent Look to your web site*

1.1 What is CSS?

The W3C invented Cascading Style Sheets (CSS) in 1996 to increase the presentational sophistication *and* the accessibility of websites, and to eliminate the browser-specific markup that threatened to fragment the emerging web. In 1997, some browsers began to support parts of CSS-1, but the standard did not become truly usable until 2001. Today, with standards-compliant browsers dominating the market, there is every reason to use CSS to remove invalid markup from your sites, separate style from content, lighten the bandwidth of your pages, and increase the odds that people and devices will actually be able to access the sites you create.

CSS gives more control over the appearance of a Web page to the page creator than to the browser designer or the viewer. With CSS, the sources of style definition for a given document element are in this order of precedence:

1. The STYLE attribute on an individual element tag
2. The STYLE element that defines a specific style sheet containing style declarations or a LINK element that links to a separate document containing the STYLE element. In a Web page, the STYLE element is placed between the TITLE statement and the BODY statement.
3. An imported style sheet, using the CSS @import notation to automatically import and merge an external style sheet with the current style sheet
4. Style attributes specified by the viewer to the browser
5. The default style sheet assumed by the browser
6. CSS describes **how HTML elements are to be displayed on screen, paper, or in other media**
7. CSS **saves a lot of work**. It can control the layout of multiple web pages all at once
8. External style sheets are stored in CSS files

1.2 Defining a Style

A style is defined with two sections: one or more selectors and one or more declarations enclosed in {Curly Brackets. Each declaration is made up of a Property: Value; pair.

```
p {
font-size: 10pt
}
```

- The selector in this example is redefining the P HTML element to have a size of 10 points
- Brackets must be curly brackets.
- The colon : separates the property from the value.

Using more than one Selector

h1, h2, p {color: #000033}

The elements h1, h2 and p will all be coloured blue

Using more than one Declaration

We could use the following

```
p {font-size: 10pt;}
p {color: green;}
```

OR a shorter way is:

***selector* {*property*: *value*; *property*: *value*; }**

```
p {font-size: 10pt; color: green; }
```

Example:

```
 body
{
    font-family: Verdana, Arial, Helvetica, sans-serif;
    font-size: 14px;
    background-color: #3333FF;
    color: #000033;
}
```

- The semi colon ; is used as a separator.

- The code can be written all on 1 line, white space has been used in the example to improve readability.

1.3 Types of Style Sheets

There are 3 types of style sheets. The difference is where the style is defined and the area where that style is applied.

- **Inline style sheet** This type of CSS is defined within a tag. Applies only to that particular occurrence of that tag. Inline style sheet are should only be used where a particular style is not going to be repeated elsewhere on the page/site. The definition is defined within the HTML tag in the body section of the HTML code. It must be redefined every time it is required. The sytax for inline styles is slightly simpler than that of Internal and External styles in that there is no selector and no curly brackets.

<element STYLE="property:value">

Using style example

<Body STYLE="COLOR : RED">

This example will change the Text color inside the BODY tag to RED

- **Internal (also called Embedded) style sheet** is defined within the head section of a page. Applies to that page only. The definition is written once in the head part of a page. It must be written on every page that requires that style. Because of the above it is ideal if only 1 page is going to be used for this style. The styles can then be used more than once throughout the page.

The Internal style sheet is defined within the head section.

<head>
< style type="text/css">
Your Style definitions go here
< /style>
< /head>

Example:

```
1  <%@ Master Language="C#" AutoEventWireup="true"
2      CodeFile="MasterPage.master.cs"
3                      Inherits="MasterPage" %>
4  <!DOCTYPE html>
5  <html xmlns="http://www.w3.org/1999/xhtml">
6  <head runat="server">
7  <style>
8      .body {
9      background-color: gray;
10     color:red;
11     }
12     .button {
13         font-family:Verdana;
14         font-size:12pt;
15     }
16 </style>
17      <title>:: Students Portal :: </title>
18 </head>
19 <body>
20     <form id="form1" runat="server">
```

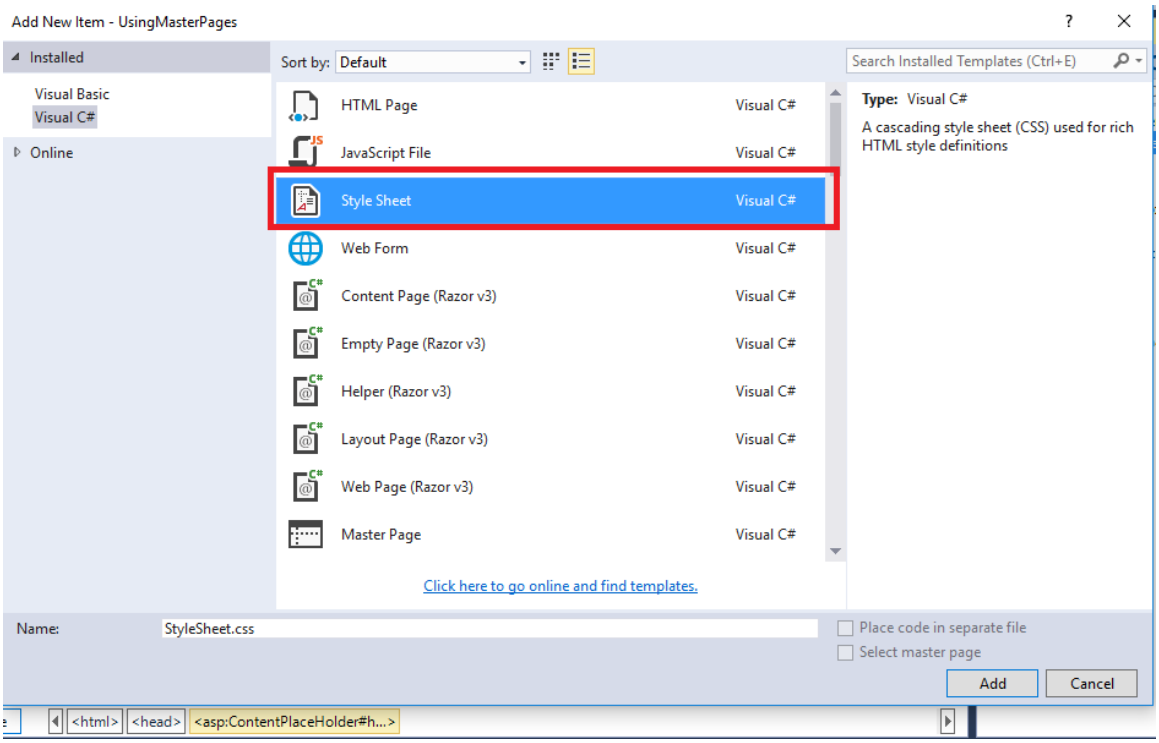
- **External style sheet** defined in a separate, hence external, file. Applies to all pages that link to the external style sheet. Separate style sheet file is created for External style Sheet. This is the best method if you wish to control the design of more than one page. The style definitions are only written once and saved into a file. Each page that wishes to use that file places a link to the file in the HEAD section. <head> Section of your pages. Link to the Style file with the <link> tag. Place the following link into the <head> section of your page, use the name you have selected for the CSS file in place of yourStyleFileName.css

<link rel="STYLESHEET" href="yourStyleFileName.css" type="text/css">

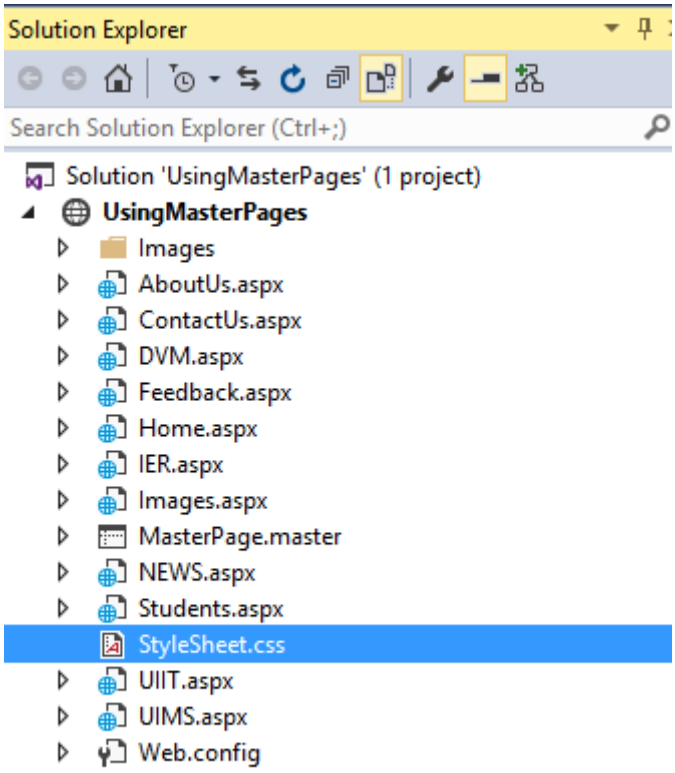
Note: The link tag does not have a closing tag in HTML

Follow these steps to define the External CSS.

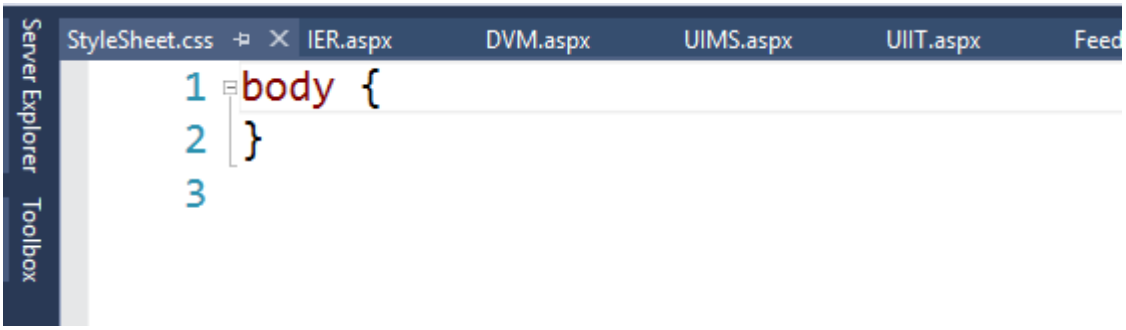
1. Right Click on the **Web Site Name,** Choose Add ➔ Add New Item ➔ Choose StyleSheet file



2. Press the Add button, you will see that StyleSheet.CSS file is added to your web site in Solution Explorer



3. Its default look is as



4. Add some CSS classes to this file e.g. Button, TextBox, Page Heading etc. Each CSS Class name must be preceded by **Dot (.)**

```
body {
}
.PageHeading {

font-size : 16pt;
font-family:Verdana;
color :  #621d1d;
}
.button {
font-size:12pt;
font-family:Verdana;
font-weight:bold;
}
.textbox {
font-size:10pt;
font-family:Verdana;
font-weight:bold;
color : red;
}
```

5. Here, we have **Master Page** defined for our web site. When Master Page is defined for the Web Site, then we need only to add the External CSS link to Master Page and each of the class is accessed in any of the Web Form to which Master page is applied. Drag and drop the StyleSheet.CSS to the head section of the Master Page as shown

```
1  <%@ Master Language="C#" AutoEventWireup="true"
2      CodeFile="MasterPage.master.cs"
3                          Inherits="MasterPage" %>
4  <!DOCTYPE html>
5  <html xmlns="http://www.w3.org/1999/xhtml">
6  <head runat="server">
7      <link href="StyleSheet.css" rel="stylesheet" />
8       <title>:: Students Portal :: </title>
9      <asp:ContentPlaceHolder id="head" runat="server">
10     </asp:ContentPlaceHolder>
11 </head>
12 <body>
13     <form id="form1" runat="server">
14         <table width="100%" border="2">
```

6. Next, open the Home Page and Apply the PageHeading CSS to the Page Title as shown

```
Home.aspx*   MasterPage.master*
1  <%@ Page Title="" Language="C#" MasterPageFile="~/MasterPage.master"
2              AutoEventWireup="true" CodeFile="Home.aspx.cs" Inherits="Home" %>
3
4  <asp:Content ID="Content1" ContentPlaceHolderID="head" Runat="Server">
5  </asp:Content>
6
7  <asp:Content ID="Content2" ContentPlaceHolderID="ContentPlaceHolder1"
8                 Runat="Server">
9      <table>
10         <tr>
11             <td class="PageHeading"> :: Home Page :: <hr /></td>
12         </tr>
13     </table>
14 </asp:Content>
```

7. Save and Browse The Web Site. You will see that CSS class PageHeading is applied to the Page Heading text using <TD> tag.
8. Similarly, you can apply the Button, TextBox and other CSS classes defined in the external CSS file to different controls on Web Form.

Note that more than one of these 3 types can be used on a page, and you can link more than one External style sheet to a page.

1.4 Cascading Order of Precedence

The term Cascading is used because more than 1 style sheet can be used on any particular page. If the same style is defined with different **values** in the different style sheets then the order of precedence is

Inline Style Sheets

Internal Style Sheets

The **last External Style sheet** moving in order to the **first External Style Sheet** linked in the head section,

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